Introduction

Task analysis stands as a pivotal element in crafting user interfaces that resonate with the intended audience. This also extends to the aero plane industry, where our focus lies in catering to the preferences of millennials and Gen X. Our goal is to usher in a new era, eliminating the challenge of carry-on unavailability and, consequently, reducing flight delays. Centering on the user travel experience, this task analysis seeks to identify essential technical considerations for the interface. Through a curated set of questions, users will have the opportunity to tailor the interface, ensuring a seamless and personalized interaction.

Interface Proposal

The current chaos surrounding carry-on spaces has led to countless flight delays, with reports indicating 1.6 million delayed flights from Jan 2019 to Feb 2023. One contributing factor is the unavailability of carry-on spaces, forcing carriers to check these bags, causing delays that range from minutes to hours. Our app aims to tackle this issue head-on, offering users a seamless flight experience and reducing the workload for flight attendants. It will be available on the Apple Store and Google Play Store for easy access and security, minimizing the risks associated with website visits.

Upon launching the app, users will encounter a straightforward login page for secure access. The page offers two options: login or sign up, following NIST guidelines for robust password security during sign-up. Once logged in, users are greeted by a welcome page displaying their information and a quick overview of flight details. Flight status checks, timely notifications for delays or cancellations, and announcements on promotions or maintenance will be provided. A quick action toolbar and navigation bar streamline user access to essential features, such as reserving a spot, airline information, rules, regulations, and contact details for assistance. The footer ensures users are aware of airline terms and conditions.

Delving into the reservation page, users will find a list of upcoming flights, automatically populated through airline collaboration or manually entered if needed. For inaccuracies, a help desk pop-up will appear allowing the user to seek assistance immediately. The app then displays a plane map, highlighting the user's seat and color-coded carry-on spaces (unavailable in gray, available in green, and taken in red). Users select a space, confirm details in a pop-up, and proceed to payment using various options. A QR code is sent for boarding verification.

This app interface prioritizes technical requirements for a user-friendly experience, offering resources for convenient carry-on planning. The personalized approach considers user requirements and feedback for an enhanced travel experience.

Goals of Task Analysis

1. How can we analyze these tasks to gain a more comprehensive understanding of user experience?
2. What are the Issues and their respective complexity do users encounter when dealing with carry-on spaces?
3. How can the task analysis contribute to enhancing the efficiency of reserving a spot and reduce delays?
4. How can the task analysis be used to minimize workload for flight attendants and crew members?
5. How can we implement a secure login process and protect the information of a user?
6. How can we make this app easy to use for the users with no experience with technology?
7. How can we add text to speech for users with certain disabilities?
8. How can the app encourage users to plan early for their flights by providing early information / incentives?
9. How can task analysis help in establishing effective communication channels with users, such as notifications for flight status, announcements, and contact information?
10. How will task analysis ensure that the app interface guides users to adhere to airline terms, conditions, and regulations, preventing unintentional violations?
11. In what ways can task analysis contribute to empowering users to customize their interface based on their preferences and needs?

User Population

People between the ages of 18-55 tend to travel the most, so we want to create an app that simplifies the journey of flying and with the creation of an app to reserve an overhead bin for your carry-on bag it will make the process of finding a spot easier for anyone that struggles with getting their carry-on bag in the bin.

Detailed Task Analysis Plan and Justification

When thinking about the interface of our app, we need to understand how the users will navigate it and we need to think about how we're going to start planning our app. Since there aren't that many apps that deal with reserving a carry-on spot, we have to think about us, the creators, as the user and we also need to interview other people so that we can get a second opinion so that we can have accurate information when creating the interface of the app. We want anyone in the age group 18-55 to use the app and we must

understand that some people aren't as internet savvy as most people. Since we understand that some users will have less experience in using apps, we have to make the interface easy to read as possible since that way users can understand the purpose of our app and not have to worry about it being too evolved so that only users with app experience can utilize the app. The app will be free to download but a membership will be $15 since we didn't want the price to be too high and have customers not use the app if it was a high cost.

Understanding user's issues when trying to get an overhead bin is essential since the point of our app is to simplify the process since you already have a reserved spot before you board the flight, resolving the issues that would occur if the user hadn't reserved a spot. By understanding customer's frustrations on how airlines operate with the procedure of carry-on bags, we will get a good idea on what to include in the application since the surveys will have real user feedback, which is crucial to any application. We chose to create a series of questions, in survey style, that can help us gather crucial information we need to start the app since we need to know if the user has experience flying in an airplane and if they do or don't, we need to know how extensive that experience is because it can differ between users since some are frequent flyers while others are not. Since this app will be based on the user's needs, we made sure to ask questions that were going to help us with the interface of the app since we need to know what customers are looking for in an app that reserves a carry-on space overhead.

Wrap-Up

By creating the questionnaire , we hope to gather good and accurate feedback to start creating our app. By implementing this questionnaire, it is a good way to collect data from people of all ages and that's one of the key elements in our app since we're not focused on just one particular age group. By creating the questions in a survey style, we are going to benefit since we can just send it out to tons of people and they can respond at their convenience. The questions allows us, as the creators, to understand if users are frequent flyers or not, if they have any worries about security on the plane, if they are comfortable using technology, or the flight status and whether they would want to be notified about that. By collecting that data, it really helps us since we need that user feedback to accurately make our interface so that we can hit our goals and make an app that hopefully anyone can use. By using the feedback they give us, we can create features that users would like to see in our app.

Conclusion

In the task analysis, we went through the plans and goals that we want to utilize to create our app and make sure our users have a good experience when interacting with our app. We created a survey of questions because we want to make sure we have a good

base to start off with so that we can go in the right direction and consider users feedback when creating the interface of the app. We listed our goals that we want to achieve in order for customers to have a good experience. We want to know how to make our app align with our customers needs and that is an important thing when creating an app. We want new users to use our app and recurring users to keep coming back because that is how we know if our application is good. With the plans and goals in mind, we hope to create an app that helps users worry less about the process of securing a spot in the overhead bins for their carry-on, hopefully making our app the first thing users think of when wanting to secure a spot on the plane.

Questionnaire

Age:

Gender:

1. How frequently do you travel by air
    - ☐ Once a month
    - ☐ 2-3 times a month
    - ☐ 4 or more
    - ☐ I do not travel
2. Do you travel for business or leisure
    - ☐ Business
    - ☐ Leisure
3. Have you ever experienced delays or issues related to carry-on spaces during air travel
    - ☐ Yes
    - ☐ No
4. Have you ever experienced delays or issues related to carry-on spaces during air travel?
    - ☐ Yes
    - ☐ No
5. Do you think carry-on reservations should exist?
    - ☐ Yes
    - ☐ No
6. How comfortable are you with using technology when it comes to traveling
    - ☐ Expert
    - ☐ Proficient
    - ☐ Adequate

- ☐ Inexperienced

7. What are some of the features you consider essential in an app designed for reserving carry-on spaces?
8. How concerned are you about the security of your personal information when using a travel app?
   - ☐ Very Concerned
   - ☐ Not as Concerned
   - ☐ Not Concerned
9. What factors influence your decision when reserving carry-on spaces?
   - ☐ Cost
   - ☐ Time
   - ☐ Availability
   - ☐ Other:
10. How would you like to be notified about your flight status and any potential delays?
    - ☐ Text
    - ☐ Email
    - ☐ Other:
11. How likely are you to use an app that offers promotions or incentives for early carry-on reservations?
    - ☐ Very likely
    - ☐ Less Likely
12. How important is cross-platform accessibility to you when choosing a travel app?
    - ☐ Very important
    - ☐ Not as important
    - ☐ Unimportant
13. In what ways would you like to personalize your experience with a travel reservation app?
14. What channels do you prefer for seeking support or assistance within a travel app?
    - ☐ Phone
    - ☐ Live chat
    - ☐ Email
    - ☐ Other:
15. How satisfied are you with the current system of getting a carry on spot?
    - ☐ Very satisfied
    - ☐ Satisfied
    - ☐ Average
    - ☐ Unsatisfied

16. What improvements would you bring to make this experience more satisfactory?